

MUGILAN S

Electrical & Electronic Device Specialist | Technical Writer
Mechanical structural Design | Front-End Developer | UI/UX Designer



ABOUT ME

I AM MUGILAN, ELECTRONICS AND COMMUNICATION ENGINEER WITH HANDS-ON EXPERIENCE IN EMBEDDED SYSTEMS, IOT, AND APPLIED ENGINEERING SOLUTIONS. CURRENTLY DEVELOPING HARDWARE AND SOFTWARE SYSTEMS FOR INDUSTRIAL AND DEFENCE-ORIENTED APPLICATIONS, INCLUDING AIR PURIFICATION AND ENERGY SYSTEMS. STRONG BACKGROUND IN TRANSLATING COMPLEX ENGINEERING CONCEPTS INTO CLEAR, STRUCTURED, AND USER-FRIENDLY DOCUMENTATION FOR TECHNICAL AND NON-TECHNICAL AUDIENCES. PROVEN ABILITY TO LEAD MULTIDISCIPLINARY PROJECTS FROM CONCEPT DEVELOPMENT THROUGH PROTOTYPE IMPLEMENTATION, TESTING, AND TECHNICAL REPORTING.

CONTACT ME

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EDUCATION

sri Aurobindo Mira

Matriculation Higher Secondary School
2007-2022

k.L.N.College Of Engineering
Electronics and Communication
Engineering

2022-2026

TECHNICAL SKILLS:

- Embedded Systems
- Verilog HDL
- Electric Vehicle Design
- UI/UX Design
- Adobe Design
- Microsoft
- AI Prompt Development
- Video Editing

PERSONALITY TRAITS

- Detail Oriented
- Strong Research & Writing Skills
- Adaptable & Quick Learner
- Team Collaborator
- Problem Solver

CERTIFICATES

- DIPLOMA IN ELECTRIC VEHICLE DESIGN
- DIPLOMA IN CORPORATE ENTREPRENEURSHIP
- UI UX
- EMBEDDED SYSTEM INTERNSHIP CERTIFICATE
- ANALOG DESIGN CERTIFICATE
- VERILOG HDL CERTIFICATE
- PCB BOARD DESIGN

KEY STRENGTHS

- **Simplifying Complexity** – Transform complex technical concepts into clear, concise, and easy-to-understand content for diverse audiences.
- **User-Centric Writing** – Develop content with the reader's needs in mind, ensuring clarity, usability, and engagement.
- **Structured Documentation** – Create well-organized manuals, guides, and technical documents with logical flow and actionable instructions.
- **Accuracy & Technical Reliability** – Leverage engineering expertise to produce precise, fact-checked, and technically sound documentation.

AWARDS

• Design Patent Granted – “CO2 Purus” (Aug 2025)

Secured a design patent for an air purification system concept

• 2nd Prize – Build Expo 2025 (Aug 2025)

Won second place among multiple project entries; awarded ₹4,000

• Community Innovation Fellow (CIF) – Atal Innovation Mission, NITI Aayog (Sep 2025)

Selected for national-level innovation program with ₹2,00,000 grant support

• National-Level Innovator – Bharat Innovators Program (Dec 2025)

Recognized for innovation and contribution to technology development

EXPERIENCE

Founder & Lead Engineer

- AWLRI Group of Research
- 2024 – Present
- Lead design and development of embedded systems and IoT-based engineering solutions
- Built prototypes for air purification, smart agriculture, and waste-to-energy systems
- Integrated ESP32, sensors, and automation systems for real-time monitoring and control

Developed both hardware and software components for industrial applications

Defence Technology Collaborator

- In collaboration with Rudram Dynamics Foundation
- 2025 – Present
- Contributing to development of defence-related technology solutions
- Involved in prototype development, testing, and system-level integration
- Supporting field trial preparation and deployment activities

AFFILIATION

Associated with Atal Innovation Mission, NITI Aayog

PROJECTS

• FUELECTRIC POWERPLANT (UNDER RESEARCH AND DEVELOPMENT PROCESS)

A fuel-based electric power plant generates electricity by burning PLASTIC which drives a turbine connected to a generator.

• CO2 PURUS (UNDER DESIGN PATENT PROCESS)

CO2 Purus is a device designed to monitor and reduce carbon dioxide (CO₂) levels in indoor environments, ensuring better air quality and a healthier atmosphere through advanced filtration or absorption technology.

• MONOPURE (UNDER PATENT PROCESS)

Tox Pure Aero is an advanced air purification system designed to remove toxins, pollutants, and airborne contaminants, ensuring clean and breathable air for a healthier indoor environment.

• KUPBIN (UNDER RESEARCH AND DEVELOPMENT PROCESS)

KUPBIN is a smart IoT-based waste management system that detects waste levels, sends real-time alerts, and enables remote waste compression to optimize bin capacity and collection efficiency.